Md. Abdullah Al-Fahim

Civil Engineering Researcher | Computational Geomechanics

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RESEARCH SUMMARY

Civil engineer working between experimental soil mechanics and computational modelling. I build **discrete element models** of granular soils from scratch and calibrate them against matched laboratory tests, currently on how the inherent (bedding-plane) anisotropy of sand carries over to sand–structure interfaces. I am seeking a graduate (**MSc**/**PhD**) research position in computational geomechanics and related areas.

Interests: discrete element method; inherent anisotropy and fabric; soil–structure interfaces; contact-model calibration and validation; specimen-size effects in direct shear; sustainable construction.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) **2019 – 2024**

Bachelor of Science in Civil Engineering *Rajshahi, Bangladesh*

Thesis: *"Influence of Inherent Anisotropy of Soil on the Behavior of Sand-Concrete Interface"*
Supervisor: Prof. Md. Mahmud Sazzad

RESEARCH EXPERIENCE

Discrete Element Modelling of Sand and Sand–Concrete Interfaces **2023 – 2024**

Undergraduate thesis research *Dept. of Civil Engineering, RUET*

- Developed a custom two-dimensional, **GPU-accelerated Discrete Element Method (DEM) engine** from scratch in Python/PyTorch (float64) to simulate direct shear of clumped-grain assemblies, using elastic–perfectly plastic tangential and rolling-resistance contact laws and a sawtooth concrete interface.
- Designed a matched laboratory programme of **48 direct shear tests** (ASTM D3080) on a river sand and a crushed sand, sheared against themselves and against concrete, at four bedding angles and three normal stresses, with three random-seed replicates per simulated condition.
- Applied a strict **calibrate-once protocol**: contact parameters tuned against experiment at a single bedding angle and then frozen, so that strength at all other bedding angles is a blind prediction rather than a re-fit.
- Analysed how inherent (bedding-plane) anisotropy is transmitted from sand–sand shearing to the sand–concrete interface, combining stress–strain response with contact-normal, force-weighted and grain-orientation fabric tensors.
- Studied the effect of specimen size on whether bulk anisotropy is captured in direct shear, and used the outcome to select the specimen geometry for the interface simulations.
- Journal manuscript based on this work is **in preparation**; next, extending the framework to sand–pile interfaces under lateral loading.

Barriers to Green Buildings in Bangladesh **2024**

Survey and Analytic Hierarchy Process *RUET*

- Survey and Analytic Hierarchy Process analysis of the hindrances to green building construction (n = 87 responses; consistency ratio 0.011); **government involvement ranked first (25.0%)**. First-author conference paper (ICCESD 2024).

PUBLICATIONS

Conference Papers

- **Al-Fahim, M.A.**, Shahriar, S. (2024). *Hindrances in Constructing Green Buildings in Bangladesh: A Comprehensive Study.* 7th International Conference on Civil Engineering for Sustainable Development (**ICCESD 2024**), KUET, Khulna, Bangladesh.

In Preparation

- **Al-Fahim, M.A.**, Sazzad, M.M., Shafi, M.S.M. A discrete element study of bedding-plane anisotropy at sand and sand–concrete interfaces.

PROFESSIONAL EXPERIENCE

RANCON Engineering & Consultants Ltd. **Nov. 2025 – Present**

Executive (QC-PMO) *Dhaka, Bangladesh*

- Inspect structural, architectural and finishing works and incoming materials against approved drawings, specifications, BOQ and BNBC; resolve non-conformities and maintain QA/QC records.
- Developed a **portfolio-tracking dashboard** (Python/Flask) integrating project schedules with progress and reporting across concurrent developments.
- Developed and piloted a **web-based digital QC inspection system** with checklists, a raise–respond–verify–close NCR workflow, role-based approvals, attachments and notifications.

Mir Cement Ltd. **Jan. 2025 – Nov. 2025**

Engineer (Civil) *Narayanganj, Bangladesh*

- Developed and monitored project schedules and prepared cost estimates, budgets and resource plans.
- Designed and optimised concrete mixes per ACI 211.R and verified raw materials and finished concrete against quality standards.

TECHNICAL SKILLS

Computational: Python (PyTorch, custom DEM/FEM code), R, ABAQUS

Laboratory: direct shear testing (ASTM D3080), particle-size analysis, concrete mix design (ACI 211), relative density testing, aggregate property testing (ACV, LA abrasion, Fineness Modulus, Specific Gravity)

Methods: calibration and validation of numerical models, survey design, Analytic Hierarchy Process

Design & Productivity: AutoCAD (2D), SketchUp, Lumion; MS Word, PowerPoint, Excel

PRESENTATIONS & ACTIVITIES

- Poster Presentation – Retrofitting Old Toilets for Sustainable Bangladesh (**WaterAid Toilet Conference, 2025**)
- Team Event – Game of Trusses (**Ceenovation, IUT, 2024**)
- Poster Presentation – Sustainable Microgrids: Enhancing Campus Energy Resilience (**Technofair, RUET, 2023**)

REFERENCES

Available upon request.